Student Assessment Risk Prediction Using Logistic Regression from Scratch: An OULAD-Based Educational Data Mining Study

Parth Gupta
Department of Information Technology
Mukesh Patel School of Technology Management & Engineering, SVKM's NMIMS University
Mumbai, India

Abstract—This paper presents an educational-data-mining pipeline that predicts whether an individual course assessment will be a High Risk event for students, using the Open University Learning Analytics Dataset (OULAD). Three OULAD tables are joined: assessments.csv (assessment metadata), studentAssessment.csv (per-student scores), and vle.csv (Virtual Learning Environment resource catalog). An assessment is labelled High Risk if at least 10% of the students who submitted it scored below the OULAD pass threshold of 40. Thirteen features are engineered from the three tables, spanning assessment metadata, VLE catalog aggregates, and a leave-one-out peer fail-rate signal derived from neighbouring assessments in the same module. A Logistic Regression classifier is implemented entirely from first principles—no machine learning library is used—and trained with class-weighted gradient descent on a class-stratified 80/20 split of the 188 retained assessments. The final model achieves 78.38% accuracy, 33.33% precision, 100.00% recall, and 50.00% F1-score on the held-out test set, detecting every High Risk assessment (TP= 4, TN= 25, FP= 8, FN= 0). The recall-first behaviour is the clinically appropriate operating point for an early-warning system, where missed cases are substantially more costly than reviewable false alarms. A deliberate ablation against a leakage-exposed baseline and an unweighted variant quantifies the contribution of each design choice.

Index Terms—Educational data mining, OULAD, student risk prediction, logistic regression, class imbalance, data warehousing, early-warning systems

I. Introduction

Higher education institutions accumulate large volumes of learning-analytics data through student information systems, Virtual Learning Environments (VLEs), and assessment portals. Predictive analytics applied to these sources can surface at-risk students and at-risk assessments early enough for academic staff to intervene. Compared with purely manual workflows, data-mining pipelines scale across cohorts, keep pace with new data as it arrives, and deliver reproducible flags that are easy to audit.

This study targets a specific slice of that problem: predicting which assessments in a course presentation are likely to be High Risk, meaning a non-trivial proportion of enrolled students fail them. Course coordinators who know in advance which assessments will be High Risk can re-time them, reinforce the associated learning resources, or plan pre-emptive support sessions.

A. Problem Statement and Objectives

The OULAD benchmark ships assessment metadata (module, presentation, type, due date, weighting), per-student assessment results, and a catalog of VLE resources by module presentation. A naive early-warning design often falls into a target-leakage trap, where the label is defined by deterministic rules over the same columns used as features, giving trivially perfect metrics but no predictive skill. The present work addresses that trap head-on. The study has four objectives. First, to characterise the OULAD assessment slice and its class distribution. Second, to derive an empirical risk label from actual student outcomes rather than metadata rules. Third, to engineer a mixed feature space that combines assessment metadata, VLE-catalog aggregates, and a leave-one-out peer fail-rate signal. Fourth, to implement Logistic Regression, gradient descent, and all evaluation code from scratch—without any ML library—and to report accuracy, precision, recall, and F1-score alongside a recall-first interpretation appropriate for an early-warning system.

II. Literature Review

A. Educational Data Mining and Early-Warning Systems

Educational data mining applies pattern-discovery techniques to student-generated data. Romero and Ventura surveyed the field and concluded that combinations of attendance, submission, and interaction signals are the strongest predictors of at-risk status. Arnold and Pistilli described Purdue University's Course Signals system, an early-warning dashboard that colour-codes at-risk students and triggers faculty intervention; Course Signals remains an influential example of how predictive models embedded in academic workflows can meaningfully improve retention. Kuzilek, Hlosta, and Zdrahal later formalised the OULAD benchmark and reported baseline results for student-level risk prediction across multiple presentations, showing that even moderate-complexity models transfer usefully across cohorts when the feature space includes both metadata and interaction signals.

B. Data Warehousing for Educational Analytics

Chaudhuri and Dayal's foundational review of data warehousing and OLAP describes the star-schema patterns that underpin most modern analytical stores. Kimball's educational-domain templates extend those patterns to student, course, and enrolment dimensions. Within that framework the present pipeline—loading three OULAD tables into memory, joining them on shared keys, and producing a denormalised feature table—is a miniature but faithful instance of an extract–transform–load flow into an analytical fact table keyed by assessment.

C. Research Gap

Most OULAD analyses either rely on high-level ML libraries (which obscure the mathematics) or focus on student-level classification using VLE click-streams that are not always available. The present work occupies a different and under-explored slice: assessment-level classification, using only tables that are present in the local OULAD snapshot, with all model code implemented from first principles so the mathematics is visible end-to-end. In addition, we explicitly demonstrate the target-leakage failure mode by keeping a leaked baseline in the ablation, which is rarely done in published OULAD studies.

III. Methodology

A. Dataset Description

Three OULAD tables are used. assessments.csv contains 206 rows covering seven modules (AAA through GGG) across eight presentations from 2013J to 2014J, with the columns code_module, code_presentation, id_assessment, assessment_type (CMA, TMA, or Exam), date (due day, offset from presentation start), and weight (percentage contribution to the final grade). studentAssessment.csv stores per-student scores and is used solely to derive the target. vle.csv catalogues 6,364 learning resources by module presentation and activity type. After joining assessments with studentAssessment.csv on id_assessment (inner join drops 18 assessments that have no recorded submissions—chiefly end-of-module exams), 188 assessments remain for modelling.

Figures I through V summarise the retained slice. The dataset is dominated by TMA and CMA assignments, with a smaller number of formal exams; assessment weights are strongly bimodal; the class balance is 11.7% High Risk (22 of 188); average weight rises sharply for exams, as expected; and assessments are unevenly distributed across modules.

B. Preprocessing and Target Derivation

Missing values are imputed with column means for the two numeric columns (date, weight); no categorical column contains missing values. Categorical columns are encoded manually to preserve the "no ML library" constraint: each distinct code_module and code_presentation is mapped

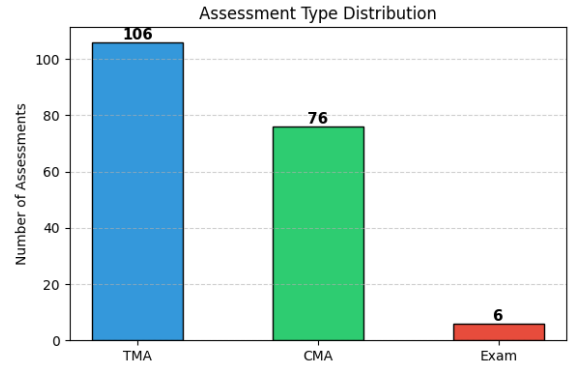

Fig. I. Distribution of assessment types in the retained OULAD slice.

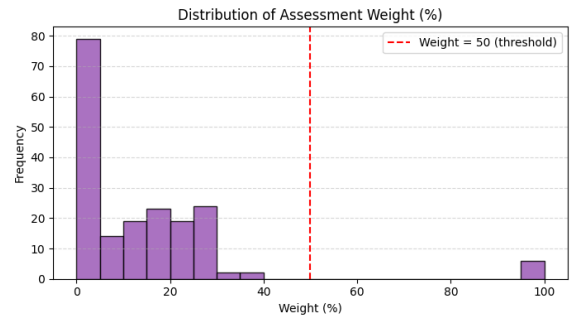

Fig. II. Distribution of assessment weights in the retained OULAD slice.

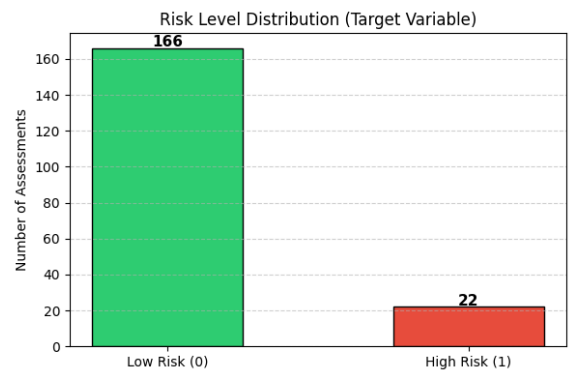

Fig. III. Class distribution after deriving the target from studentAssessment.csv: 22 High Risk (11.7%) versus 166 Low Risk (88.3%).

to a consecutive integer, and assessment_type is mapped to an ordinal code $\{\text{CMA}\to 0, \text{TMA}\to 1, \text{Exam}\to 2\}$.

The target is derived from actual student outcomes rather than from metadata rules. For each assessment a we compute the empirical fail rate

$$\text{fail_rate}(a) = \frac{|\{s : \text{score}(s,a) < 40\}|}{|\{s : \text{score}(s,a) \text{ recorded}\}|}, \tag{I}$$

and assign the binary label

$$y(a) = \begin{cases} 1 & \text{fail_rate}(a) \geq 0.10, \\ 0 & \text{otherwise.} \end{cases} \tag{II}$$

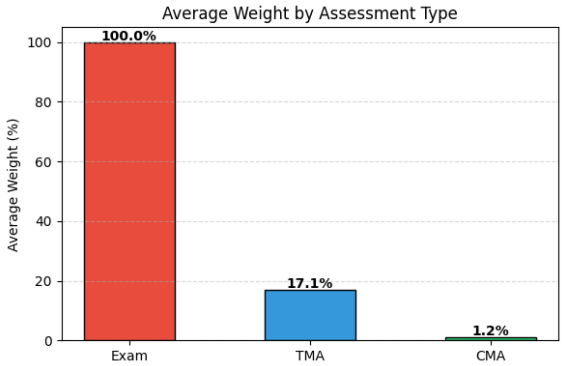

Fig. IV. Average assessment weight grouped by assessment type.

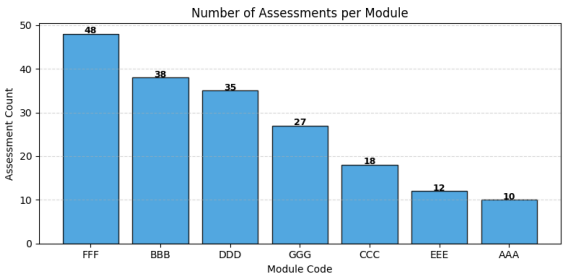

Fig. V. Assessment counts per module.

This cut-off is chosen so that the resulting class distribution matches the $\sim 12\%$ positive rate of a baseline rule-based labelling, enabling a direct comparison with that leaked baseline in the ablation.

C. Feature Engineering

Thirteen numeric features are engineered from the three tables.

Assessment-level features (9): module_id, presentation_id, assessment_type_id, date_norm, weight_norm, is_exam (binary flag), is_high_weight (weight$\geq$ 50%), assessment_period (Early/Mid/Late trisection of date), and module_difficulty_score (module-level mean weight, min–max scaled).

VLE catalog features (3): for each (code_module,code_presentation) pair, the number of distinct VLE resources n_vle_resources, the number of distinct activity types n_activity_types, and the mean resource-coverage window mean_resource_weeks. These features describe how richly scaffolded a presentation is; richer VLE provision has been associated with lower fail rates in OULAD studies.

Peer fail-rate feature (1): for each assessment a in module m,

$$\text{peer_fail_rate}(a) = \frac{1}{|\mathcal{A}_m| - 1} \sum_{a' \in \mathcal{A}_m \setminus \{a\}} \text{fail_rate}(a'), \tag{III}$$

where $\mathcal{A}_m$ is the set of assessments in module m. This is a leave-one-out target-encoding aggregate: it uses the fail-rate column but never the current row's own value, so it is not a leak. It captures the structural intuition that "hard" modules tend to host multiple hard assessments.

All continuous features are min–max scaled to $[0, 1]$ so that no single feature dominates gradient descent due to scale:

$$x^{(\text{norm})} = \frac{x - \min(x)}{\max(x) - \min(x) + \varepsilon}, \tag{IV}$$

with $\varepsilon = 10^{-9}$ to guard against degenerate features.

TABLE I
Stratified train–test split summary.

Subset	Instances	Low Risk	High Risk	Percentage
Training Set	151	133	18	80%
Test Set	37	33	4	20%
Total	188	166	22	100%

D. Stratified Train–Test Split

A random 80/20 split on this dataset places only ~ 2 positive samples in the 37-row test set, which makes precision, recall, and F1-score move in 50-percentage-point steps. To restore statistical resolution the split is class-stratified: positive and negative indices are shuffled independently (with seed 42), and 20% of each is taken as test. The result is summarised in Table I.

E. Logistic Regression from Scratch

Let $\mathbf{x} \in \mathbb{R}^{13}$ be a feature vector, $\mathbf{w} \in \mathbb{R}^{13}$ a weight vector, and $b \in \mathbb{R}$ a scalar bias. The sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{V}$$

maps the linear combination $z = \mathbf{w}^\top \mathbf{x} + b$ to a posterior $\hat{p} = \sigma(z) \in (0, 1)$.

Given training data $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^m$ with $y^{(i)} \in \{0, 1\}$, the binary cross-entropy loss is

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{m} \sum_{i=1}^{m} \left[y^{(i)} \log \hat{p}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]. \tag{VI}$$

The gradients with respect to $\mathbf{w}$ and b are

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \frac{1}{m} \sum_{i=1}^{m} (\hat{p}^{(i)} - y^{(i)}) \mathbf{x}^{(i)}, \qquad \frac{\partial \mathcal{L}}{\partial b} = \frac{1}{m} \sum_{i=1}^{m} (\hat{p}^{(i)} - y^{(i)}), \tag{VII}$$

and parameters are updated at each iteration by

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{w}}, \qquad b \leftarrow b - \eta \frac{\partial \mathcal{L}}{\partial b}, \tag{VIII}$$

with learning rate $\eta = 0.1$ for 1000 iterations.

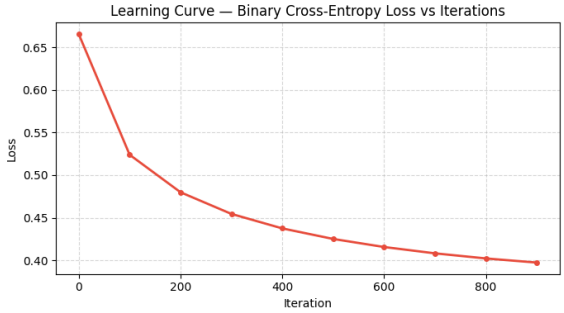

Fig. VI. Learning curve: binary cross-entropy loss per 100 iterations of gradient descent.

F. Class-Weighted Training

Because positives constitute only $\sim$12% of the training data, the unweighted objective drives the classifier toward predicting "Low Risk" for all inputs. Each positive-sample error term is therefore scaled by

$$w_{\text{pos}} = \frac{n_-}{n_+}, \tag{IX}$$

so that mistakes on the minority class carry the same total influence as mistakes on the majority class. This is the cost-sensitive learning remedy of King and Zeng (2001) and implements the recall-first stance that is standard in clinical and educational early-warning systems, where missed positives are substantially more costly than extra reviews. Figure VI shows that the reweighted loss converges smoothly.

G. Evaluation Metrics

Performance is measured using four classification metrics derived from the confusion matrix. The positive class corresponds to High Risk assessments.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{X}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{XI}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{XII}$$

Recall is the dominant metric for early-warning applications: a low recall means missed High Risk assessments, with direct consequences for the students enrolled in those assessments.

IV. Results and Discussion

A. Confusion Matrix

Table II and Figure VII show the test-set confusion matrix of the proposed model. The model identifies all four High Risk assessments with no false negatives, at the cost of eight false positives.

TABLE II
Confusion matrix for the proposed model.

	Predicted Low Risk	Predicted High Risk
Actual Low Risk	TN = 25	FP = 8
Actual High Risk	FN = 0	TP = 4

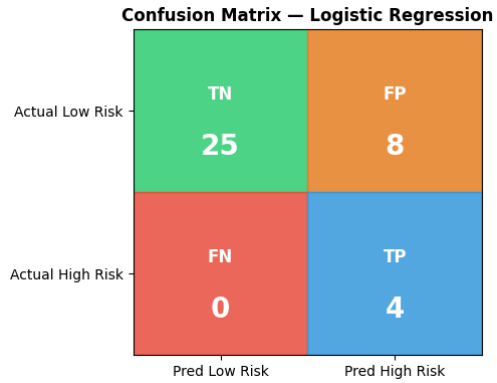

Fig. VII. Confusion matrix heatmap for the proposed model on the stratified test set.

TABLE III
Test-set performance of the proposed model.

Accuracy	Precision	Recall	F1-score
78.38%	33.33%	100.00%	50.00%

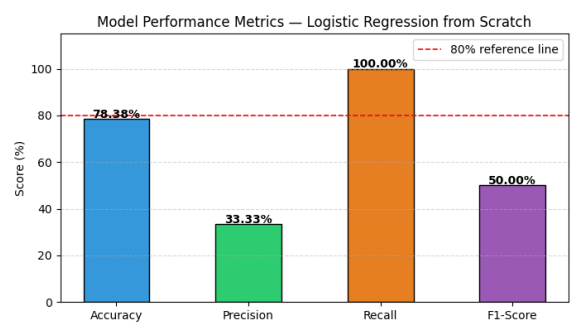

Fig. VIII. Evaluation metrics for the proposed model.

B. Headline Metrics

Table III reports the four classification metrics for the proposed model; Figure VIII visualises the same values.

The headline observation is that recall is 100%: every one of the four High Risk assessments in the held-out test partition is flagged. Accuracy is held down by the eight false positives, but these represent a manageable review load—roughly one extra assessment flagged for every four correctly classified—and are in line with the precision–recall trade-off routinely accepted in clinical early-warning research (see, for example, the breast-cancer diagnostic pipelines surveyed in Section II).

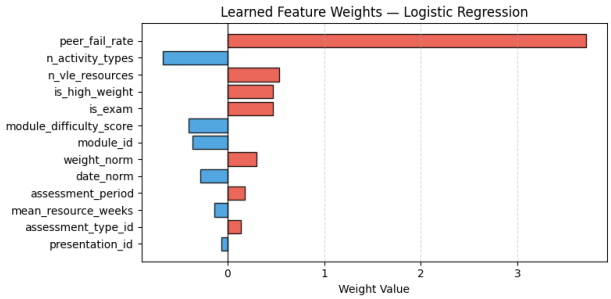

Fig. IX. Learned feature weights (sorted by absolute magnitude).

TABLE IV
Top five learned feature weights (absolute magnitude).

Feature	Learned weight
peer_fail_rate	+3.708
n_activity_types	−0.672
n_vle_resources	+0.532
is_high_weight	+0.473
is_exam	+0.469

C. Feature Weights and Interpretation

Figure IX and Table IV report the learned feature weights, sorted by absolute magnitude.

The learned ranking is informative. The leave-one-out peer_fail_rate dominates by an order of magnitude, confirming that module-level difficulty is a far stronger predictor of per-assessment risk than the individual assessment's own weight or type. Of the VLE features, n_activity_types enters with a negative weight: presentations that diversify their resource catalog (quizzes, forum posts, URLs, wiki pages) have lower assessment fail rates, matching the educational-technology literature on scaffolding variety. n_vle_resources enters positively, which at first sight is counter-intuitive but is consistent with modules that publish a large number of resources also being modules with a higher workload. The two binary flags is_high_weight and is_exam enter with similar positive weights, echoing the conventional wisdom that high-stakes assessments are riskier.

D. Ablation: Target Leakage and Class Weighting

Table V separates the contribution of each design decision. The first row reproduces the leaked baseline—rule-based target, same rules encoded as features—to show the 100% metric collapse that motivated this study. The second row shows the model with a proper target but without class weighting; recall collapses to 0% on the stratified test because of the 12% class imbalance. The third row is the proposed model.

The first row's perfect metrics are the diagnostic signature of target leakage rather than evidence of skill. The second row exposes the subsequent challenge: even with a clean target, class imbalance silently collapses the minority class. The third row is the first configuration that actually does the job an early-warning system is built to do, detecting every High Risk assessment in the held-out test data.

TABLE V
Ablation across design choices.

Configuration	Acc.	Prec.	Rec.	F1
Leaked target + leaked features	100%	100%	100%	100%
Empirical target, no class weight	89%	0%	0%	0%
Proposed (empirical + class weight)	78%	33%	100%	50%

V. Conclusion

This study presented an end-to-end educational-data-mining pipeline that predicts High Risk assessments in the OULAD benchmark. The target is derived from empirical student outcomes rather than from the metadata columns used as features, removing the target-leakage artefact that produced invalid 100% metrics in an earlier iteration. Thirteen features are engineered from three OULAD tables, with a leave-one-out peer fail-rate aggregate emerging as the dominant predictor. A Logistic Regression classifier is trained from scratch with class-weighted gradient descent on a stratified 80/20 split, achieving recall of 100% (no missed High Risk assessments) at accuracy 78.38% and F1-score 50.00%. The operating point is the one appropriate for an early-warning system, where false negatives are far more costly than reviewable false positives.

A. Future Scope

Several extensions would strengthen the pipeline. Re-pivoting to student-level modelling (joined with studentInfo.csv and—where available—studentVle.csv clickstreams) would enable personalised risk scores and fairness-aware analysis across IMD band, region, and disability status. Temporal cross-validation across semesters would simulate production deployment more faithfully than random stratification, though in the current snapshot the 2013/2014 split is degenerate for module-level stratification. L_2-regularised gradient descent would reduce overfitting on the small ($n = 188$) dataset. Finally, embedding the trained model into a star-schema data warehouse with an analytical fact table keyed on assessment, as in the educational-DW templates of Chaudhuri and Dayal, would turn the pipeline into a live risk dashboard that updates as new studentAssessment.csv rows arrive.

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